Yeast 001 — build sheet

What has to exist before this cut can be finished, ordered by what blocks what. §A is the channel's foundation and stops the graphic pipeline outright; §B is what is in the timeline today as a stand-in; §C is the shot list; §D is pickups; §E is escalated to her; §F is technical.

Nothing here is generated or approximated by the pipeline — [S-001]. Where something *is* a stand-in it says so in its filename.

Start here

1. **Import the FCPXML and watch it** (`Yeast 001 — rough cut.fcpxml`, or the preview mp4 if Final Cut isn't to hand). The cut is the thing; everything below is in service of it.
2. **Decide the three escalations that gate graphics** (E1–E3): the temperature numbers, the enzyme chain, the glutathione mechanism. Three cards are blocked on them.
3. **Scan one blank sheet** (A1). Every card in the timeline is a placeholder until it exists.

A. Blocking — the channel's foundation

Unchanged from the June 9 sheet except A3, which is done.

	What	Where	Spec
A1	**Paper grain scan**	`assets/textures/paper-grain.png`	Flat-scan a blank sheet at 1200 dpi (§4.3). Stage 5 refuses to run without it.
A2	**Her marker font**	`assets/fonts/*.woff2`	Per `font_character_set.md`. Until then the cards use **Kalam** (OFL, in `assets/fonts/`) — relaxed, marker-ish, reads as a teacher's hand; swap is one constant in `tools/cards.py`.
~~A3~~	~~Work Sans~~	`assets/fonts/WorkSans[wght].ttf`	**Done.** OFL, variable weight.
A4	**The channel LUT**	`assets/luts/yeast001-grade.cube`	**First version done** — see §G. A grade for this footage, not yet the channel's look; it will want revisiting against the T3i once that body is in rotation ([S-008]).
A5	**Foley library**	`assets/foley/`	Cardboard taps, paper slides. The placement code is written.
A6	**Cutouts**	`assets/cutouts/`	None exist yet. Every graphic in §B2 below is a cutout job.

B. What is in the timeline now

B1. Placeholder cards — 11 of them

`build/graphics/yeast001/*_placeholder.mov`, on the Graphics.Cutouts lane, at the chapter starts. Flat `bone` under Kalam, walnut ink, rust underline drawing on over 4 frames, placed on the §5.3 curve with §5.2 jitter, held per §5.4. **The timing is final; the pixels are not.** §6.1 wants a scanned surface and §10 fails a flat fill — they are named `_placeholder` so they cannot be mistaken for the card. When A1 lands, `./run tools/cards.py cuts/yeast001/picks.json` re-renders them onto it.

TC	Card	TC	Card
0:13	*Yeast* (title, 120 px)	7:25	*Access and need*
0:13	*Yeast is old*	9:30	*Budding*
1:45	*Louis Pasteur*	10:01	*What it eats*
3:09	*One bug*	10:43	*Temperature*
5:25	*What's on the shelf*	11:24	*Faster and slower*
		12:48	*The same little bug*

B2. Wikipedia pictures — 3 of them, real

`cuts/yeast001/wikipedia/` with `CREDITS.md` (source page, licence, artist for each). Placed in the left column beside her, never over her face, on the same clock as the cards. These are photographs, not generated — they don't violate [S-001] — but the flask one carries a black background that §4.1 doesn't allow in a graphic. It's a reference until the flask is drawn.

TC	Picture	Licence	Hold
1:47	Pasteur, by Nadar (1886)	Public domain	7 s
2:11	Swan-neck flask, Wellcome Collection	CC BY 4.0 — **credit required on screen or in the description**	10 s
3:35	*S. cerevisiae* SEM (Murtey & Ramasamy)	CC BY 3.0 — **credit required**	8 s

Also fetched, not placed: `yeast-dic.jpg` (budding cells under DIC, public domain) for the Budding chapter if the cutout cycle isn't ready; `pasteur-flask-glass.jpg` (CC BY 4.0), an alternate flask on a light background.

B3. Inserts from the b-roll folder — 10 split edits

TC	Clip	Under
0:00	`Best Yeast Broll` @ 4 s	"This is yeast bubbling in action"
0:02	`Katie Dancing` @ 57 s	"way too excited about yeast"
5:52	`MVI_0026` @ 3 s, 6 s	"from the big brewers … to the at-home enthusiasts"
5:32	`MVI_0027` @ 6 s	active dry
5:39	`MVI_0028` @ 1.5 s, then @ 3.4 s for 5 s	instant (two picks — a false start is cut between them)
5:51	`MVI_0031` @ 1 s, 8 s	fresh yeast
6:30	`MVI_0029` @ 2 s, 10 s	the mead packet
11:58	`MVI_0023` @ 14 s, 9 s	"this is what I have going right here"
12:25	`MVI_0025` @ 20 s, 7 s	the pre-ferment

Unused from the folder: `MVI_0022`, `MVI_0030`, `starter-jar-rise` (the 4K 30p timelapse — prepped 1080p version exists from the last cut; it was the first shot last time and could be again if the starter glass reads as too busy).

B4. Framing — 6 transforms, all editable

Two static punch-ins at 1.08× (the fungus line at 3:35, "grows where there are nutrients" at 9:50) where both neighbours run ≥ 10 s, so the cut between two takes of the same framing reads as a second camera rather than a jump. A third, on "eats through its food", was dropped when the pause-tightening left only 7 s of settled picture before it — the assembler refused it. Four slow pushes — 1.05× under the hook claim (0:05), 1.06× under "it's really all just the same" (7:14), 1.05× under "you do not actually need to add sugar" (12:17), 1.04× under the close (12:48). Each is an `<adjust-transform>` on the clip: select the clip, Transform, drag. `tools/assemble.py` refuses a static punch-in beside a pick shorter than 10 s.

C. Shot list — still wanted

The cut has a pattern interrupt inside every 40 s window except 8:08–8:38 (§9). These are the shots that would earn their keep, in order of how much they fix.

TC	Shot	Why
11:24–12:25	**Sugar vs no sugar, two jars, one clock**	The failure-then-fix the whole last chapter is about. Locked-off timelapse, 3–4 h.
9:00	**A cut loaf — stretchy crumb vs short crumb**	"More of a shorter texture … in a sandwich bread." Windowpane both doughs, then cut both.

TC	Shot	Why
0:27	**Wild yeast bloom on a dark grape**	Macro. She names it; it is also beautiful. Storyblocks if it can't be shot.
11:32	**Fat, salt, spice into a bowl**	Close, three things in, that's it.
10:37	**Dough rising / oven spring**	"The gas that we will eventually bake off." A glass-door oven and a timelapse.
4:37	**Feeding a mother**	Pairs with the comparison card. Worth shooting once from three angles — it recurs.
8:08–8:38	**A supermarket baking aisle**	The one window with no interrupt. The actual small yeast shelf lands the point.
5:25	**The whole table of jars, wide, hands**	An establishing shot for the insert run.

Graphics to build (cardboard, per `molecular_kit.md` and §1.1), in timeline order: timeline card (0:18) · the exchange (0:52) · *Selective breeding* key term (1:19) · ✗/✓ batches (1:29) · the swan-neck flask, drawn (2:11) and the dust in its bend (2:47) · *Saccharomyces cerevisiae* name card (3:10, 6:49, 12:48) · mother-vs-measured comparison (4:37) · *Access and need* key term (7:30) · the two bakers (7:43) · *Glutathione* key term (8:38, **blocked E3**) · gluten matrix (8:43) · budding cycle (9:37) · pathway (10:03, 10:25, **blocked E2**) · gauge (10:52, **blocked E1**) · end card (13:02, **blocked D1**). All of these are to-do markers in the FCPXML at the moment they belong.

D. Pickups

	What	Why
D1	**The last line, and a next action — under 20 s together**	The take's button, "*It's the same little bug, Saccharomyces cerevisiae*", is cut off by the camera: MVI_0038 stops at 12:07.81 (the 12-minute clip limit) with "cerevisiae" half said, so the cut ends on "…for at-home baking". Say the button once more, then the one thing you want them to do — §9 wants a next action and nothing on either tape has one. Same table, same cardigan.
D2	**Meyen, in the right order**	The only take has "After Louis Pasteur, we get to 1838", and 1838 is before Pasteur. Suggested: "*Twenty years before Pasteur's flask, in 1838, Franz Meyen had already given it a name: Saccharomyces cerevisiae.*" Confirm the date and the credit first (E4).
D3	*(optional)* **"because of germ theory"** without the stumble	MVI_0041 1:55. Only if it bothers you; the stumble is honest.

E. Escalated to her

The pipeline gets no vote on these (`config.ESCALATE_TO_HUMAN`). Each ends up on screen as a number or a diagram, and a graphic can't hedge the way speech can.

	Claim, as cut	Why
E1	**50 °F** dormant · **93 °F to 100-and-a-couple** sweet spot · **140 °F** killed	June 9 tape said 54 / "just over 100" / 140. Two shoots, two sets of numbers; the gauge carries one.
E2	starch → α- and β-amylase → sugars; **sucrose** → **invertase**; simple sugars → **zymase** → CO_2 + alcohol	Better than last time (no "starch → sucrose"), but she restarts five times to get there and is reading. Sign the chain before the pathway is built.
E3	Dried yeast releases **glutathione**, which weakens the gluten network and helps starch release water, so the crumb is shorter and dries faster	Real mechanism; the starch/water step and the "dries faster" claim want a source.
E4	Meyen, 1838, naming *S. cerevisiae*	Spelling (she says "Mayen"), date, and whether the credit is his alone (Cagniard-Latour and Schwann are usually in the same sentence). For D2.
E5	"Most sourdoughs are using *Saccharomyces cerevisiae*"	Stated as fact at 4:27.
E6	"It's a single-cell organism … member of the fungi family"	Fine, but it's the line under the micrograph, so it is effectively a caption.

G. The grade — applied in the preview, one click in Final Cut

The footage is flat: average saturation ~8, her face at ~125 luma against windows at 235. Two things fix it, and both are in the preview so the look can be judged now:

	What	In the preview	In Final Cut
G1	**The LUT** — midtone lift (0.5 → 0.545), highlights rolled to 0.965 so the windows never clip (§7), shadows and mids warmed (+0.03 R / −0.04 B), saturation ×1.35	`lut3d` on every a-cam clip, before the graphics	Effects ▸ Color ▸ **Custom LUT** on each a-cam clip (or on a compound of the whole storyline), pointed at `assets/luts/yeast001-grade.cube`

	What	In the preview	In Final Cut
G2	**The face lift** — a soft-light ellipse over her head and shoulders, ≈ +8 % luminance, 140 px feather, centred (960, 330)	preview-only mask, `build/roughcut/yeast001.preview.face-lift.png`	Color Board ▸ Exposure, midtones +6–8, inside a **shape mask** over the same ellipse; keyframe the mask if she leans

Measured on a table frame: face 138/126/119 → 162/143/127; kitchen 119/99/91 → 146/112/91. Peak white 235 → 228. The b-roll inserts take the LUT too — they were shot in the same rooms. The LUT is baked by `tools/lut.py` from the ffmpeg chain in `make_picks.py` (`GRADE`), so the numbers live in one place. Not in the FCPXML: a LUT effect reference is a Motion template id that can't be checked without Final Cut's own import, and a wrong one fails the whole file ([S-017]) — one click per clip beat that risk.

F. Technical

	Issue	Fix
F1	**Room tone.** 69 joins, no tone under any of them.	30 s of silence from each of the two setups. §12 check 16.
F2	**Level.** The whole programme is her unprocessed camera audio.	Expected — the §8 chain lives in the finish. The beds are measured against the *cut* voice in the 200 Hz–4 kHz band and will need re-measuring once the chain is on.
F3	**"between … 50 degrees" at 10:53** — a 1.6 s pause after "between" on tape.	Left whole rather than cut across the pause; the read-back hears the number. Listen once.
F4	**Two setups, hard cuts between them.** Table ↔ kitchen at 2:07, 3:09, 3:46, 4:28, 6:08.	Motivated (the asides are asides) but §8 wants a J- or L-cut on every setup change. Drag the audio edits a few frames either way in Final Cut.
F5	**Captions.** No `.itt` yet.	The read-back words are in `build/roughcut/yeast001.heard.txt`; captions come from Stage 5 once the plan exists.
F6	**CC BY credits.** The flask and SEM pictures require attribution.	In the description at minimum; `wikipedia/CREDITS.md` has the exact lines.

	Issue	Fix
F7	**WhisperX folds false starts into the next word.** "It's" timed at 1.0 s, "yeast" at 9.6 s, "If" at 0.46 s — each one a stumble hidden inside a token, invisible in the transcript and audible in the cut.	Six cut points in this cut are set from `silencedetect` on the audio instead (`exact_start` in `make_picks.py`). The read-back is what found them, which is the whole argument of [S-026]. Worth a rule: any pick whose first word is timed longer than ~0.4 s gets its in-point measured, not read.